

ASHOK S.

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AI & DevOps professional with nearly 10 years of expertise in automation, cloud solutions, and AI-driven application development. Specialized in implementing **Generative AI**, **LLMs**, **RAG**, and **agentic AI** solutions to build production-grade AI systems. Expert in **AWS** cloud platform, **AWS Bedrock**, and **AWS SageMaker**, with demonstrated success in modernizing legacy systems into scalable microservices architectures. Certified in **AWS** and **Databricks AI**, with recognized expertise in **prompt engineering**, AI model integration, and **TypeScript/JavaScript** development.

Core Competencies

Generative AI & LLM Integration | **Cloud Modernization & Migration** | **Enterprise Automation** | **Proactive Monitoring & Observability** | Technical Mentorship | Business Application Enhancement | **Performance Optimization** & Risk Mitigation | Cross-functional Collaboration | Continuous Learning & Upskilling | Solution Innovation

Awards & Recognition

Spirit of Chennai Award

Agile Imaginative Award for Value-Added Deliverables

Key Accomplishments

- ★ Designed and implemented **AI-assisted dashboards**, reducing monitoring blind spots by **70%**
- ★ Reduced repetitive data processing efforts by **40%** using **GenAI-powered automation** solutions
- ★ Enhanced application error detection accuracy by **75%** using **AI-driven anomaly detection** tools
- ★ Improved cloud resource utilization efficiency by **20%** through **serverless and containerized deployments**
- ★ Accelerated **monolith-to-microservices migration**, improving deployment predictability by **30%**
- ★ Optimized cross-functional team workflows, resulting in **20% improvement** in collaborative project delivery efficiency

Technical Skills

Programming Languages

TypeScript, JavaScript, Python, NodeJS, Java, Shell Scripting, SQL, PL/SQL

AI/ML Frameworks

LangChain, LangGraph, Databricks AI, CosmosAI, **ClaudeAI, Hugging Face, PyTorch**

Vector & Databases

Pinecone, Chroma, FAISS, Qdrant, **pgvector, PostgreSQL**, Oracle, MongoDB

AWS Cloud Services

Bedrock, SageMaker, Lambda, S3, EC2, CodePipeline, Comprehend, Kendra, Textract, SageMaker Data Wrangler, Guardrails, **Bedrock Knowledge Base, Bedrock Agent Runtime**, Transcribe, **Rekognition**, Lex, **Augmented AI**

DevOps & CI/CD

Jenkins, Git, Terraform, Docker, Kubernetes, Artifactory, Pipeline Automation

Monitoring Tools

Grafana, Prometheus, Kibana, **MLOps** Monitoring

Education & Certifications

MCA

Anna University

B.Sc. Computer Science

Bishop Heber College

AWS Certified Cloud Practitioner

AWS Knowledge: Serverless | AWS Well-Architected Proficient

Databricks AI Certifications

Generative AI | AI Agent | AI Security

Diploma in Innovation Management

IBM Institute, Berlin

Diploma in Sustainable Management

IBM Institute, Berlin



Professional Experience

DevOps Engineer 3

Oct 2024 – Jul 2025

ADITI

Project: PayPal (Payment & Instrument Vault)

- Architected and deployed production-grade AI chatbot leveraging **LLMs**, **RAG architecture**, **LangChain**, and **CosmosAI** for PayPal Funding Instrument Platform, processing 10K+ queries with 95% accuracy
- Built **multi-agent AI systems** using **LangGraph** and **ClaudeAI** to orchestrate complex workflows, automating payment instrument validation and fraud detection pipelines
- Implemented **vector database** solutions using **Pinecone** for semantic search and document retrieval, enabling sub-second query responses on 100K+ financial documents
- Developed **TypeScript/JavaScript** APIs integrating **LangChain agents** with backend microservices, enabling real-time AI-powered decision making in payment workflows
- Engineered **AWS Bedrock** integration for foundation model deployment, optimizing inference costs by 35% while maintaining performance SLAs
- Built **MLOps pipelines** using **AWS SageMaker** and **Docker** for continuous model training, evaluation, and deployment with automated rollback capabilities
- Designed comprehensive AI observability dashboards in **Grafana** tracking model performance metrics, token usage, latency, and accuracy across production environments
- Implemented **prompt engineering** frameworks with version control and A/B testing, improving response quality by 45% and reducing hallucinations
- Automated recurring production tasks using **GenAI**-powered scripts, reducing manual operational efforts by 40% and improving team efficiency
- Established AI governance and security protocols, including PII detection, content filtering, and compliance monitoring for financial regulations

Senior Software Engineer

Sep 2022 – Sep 2024

AgileAxis Pvt. Ltd

Project: AthenaHealth (Rules 2.0 Migration)

- Led development of AI-powered healthcare rules engine using **LangChain** and **LLMs** to automate medical billing validation, reducing processing time by 60%
- Architected **RAG-based system** for medical code classification, integrating **vector databases (Pinecone)** with healthcare knowledge bases to achieve 92% accuracy
- Built intelligent document processing pipeline using **GenAI models** to extract and classify medical claims data from unstructured documents, processing 50K+ documents daily
- Developed **TypeScript/NodeJS** microservices integrating **AI agents** with legacy healthcare systems, enabling real-time clinical decision support
- Implemented **AWS Lambda** and **AWS Bedrock** serverless architecture for AI model inference, achieving 99.9% uptime and handling 1M+ daily requests
- Created **prompt templates** library and optimization framework, standardizing AI interactions across 15+ healthcare applications
- Designed **MLOps workflows** with automated testing, model versioning, and deployment using **CI/CD pipelines** and **Docker** containers
- Led team of 8 engineers in migrating monolithic healthcare application to microservices architecture, improving scalability and deployment frequency by 10x
- Established **AI model monitoring** and performance tracking dashboards, enabling proactive identification and resolution of model drift
- Conducted rigorous code reviews and implemented AI governance best practices, ensuring HIPAA compliance and data privacy

DevOps Lead

Jan 2021 – Sep 2022

Standard Chartered

Project: ICDD (Integrated Client Due Diligence)

- Automated application release and deployment processes using **Jenkins** and **Rundeck**, increasing operational efficiency and reducing deployment time
- Managed **load-balanced environments** for multiple applications, enhancing system availability and resilience through infrastructure optimization
- Conducted **source code audits** to maintain high quality and robust application performance standards
- Streamlined deployment workflows using **CI/CD pipelines**, reducing errors and manual intervention by 50%
- Acted as **DevOps SME** and change coordinator for L3 application operations, ensuring smooth release and deployment cycles
- Automated repetitive infrastructure tasks through **Shell scripting** and **Oracle PL/SQL**, reducing manual efforts and improving SLA adherence
- Mentored and trained team members in **DevOps practices** and automation techniques, enhancing team productivity

Professional Experience (Continued)

Senior Analyst

Jan 2020 – Jan 2021

BNP Paribas

Project: Vision Client Collateral Management

- ▶ Delivered **L2/L3 DevOps operations** for collateral management systems, ensuring consistent service uptime and rapid incident resolution
- ▶ Developed **automation scripts** using **Shell** and **Oracle PL/SQL**, reducing manual operational efforts by **35%**
- ▶ Managed incidents and **post-incident analysis** processes, conducting **root cause analysis** to prevent recurrence and improve system reliability
- ▶ Coordinated **application release and deployment** activities, ensuring high availability and compliance with operational standards
- ▶ Supervised a **12-member operations team**, maintaining seamless operations and meeting SLA targets
- ▶ Successfully managed **collateral processing workflows**, improving client claim processing efficiency through automation

Senior Software Engineer

Dec 2017 – Jan 2020

HCL

Client: BNP Paribas | Projects: Money Transfer Management, Cash Claim Management, MTS Swift Tool

- ▶ Engineered and maintained **mission-critical money transfer systems**, ensuring **99.9% uptime** and seamless transaction processing
- ▶ Developed **automation frameworks** using **Shell scripting** and **Oracle PL/SQL**, reducing operational overhead and improving system efficiency
- ▶ Implemented **monitoring and alerting solutions** for production environments, enabling proactive issue detection and resolution
- ▶ Collaborated with cross-functional teams to deliver application releases, ensuring **zero-downtime deployments**
- ▶ Built **automated data processing pipelines** for financial transactions, improving processing speed and accuracy
- ▶ Led technical initiatives for **Cash Claim Management** sub-application, optimizing workflows and enhancing system performance

Software Engineer

Sep 2014 – Dec 2017

Syntel

Clients: State Street Investment Management Services, FedEx

- ▶ Developed and maintained **enterprise data warehouse operations**, ensuring accurate master, market, and reference data management
- ▶ Automated **financial data loads** and **reporting processes**, reducing processing time and improving operational efficiency
- ▶ Implemented cross-functional collaboration to streamline **data integration** and ensure seamless project delivery
- ▶ Designed and executed **hybrid automation testing frameworks** using **Selenium** and **Java**, improving testing coverage and efficiency
- ▶ Automated test environment setups and execution, reducing manual interventions and enhancing reliability
- ▶ Created dynamic object-learning tools and **performance analysis reports**, supporting informed business decision-making



Open Source AI Projects

CarbonLens AI

<https://carbonlens-ai.solutionsynth.cloud/>

Intelligent supply chain carbon footprint tracker powered by AWS AI services. Automatically processes logistics documents and provides optimization recommendations using AI/ML models.

AI Resume & Cover Letter Generator

<https://resume-generator-ai.solutionsynth.cloud/>

Intelligent career document generation platform powered by AI. Creates professional, ATS-optimized resumes and personalized, compelling cover letters tailored to job descriptions using LLMs, natural language processing, and advanced GenAI with prompt engineering techniques.

AI Content Studio

<https://ai-content-studio.solutionsynth.cloud/>

Serverless AI-powered content generation platform that transforms ideas into professional, SEO-optimized articles and web content using multi-agent workflow system.

Profilia

<https://profilia.solutionsynth.cloud/>

Professional online portfolio creation platform. Create stunning portfolios to showcase your work and attract opportunities.

Synth AI

<https://solutionsynth.cloud/>

Comprehensive AI solutions platform powered by AWS services and modern AI frameworks.